

The empirical recurrence for OEIS A231995, recovered from its tail

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8 September 2026

Abstract

OEIS A231995 records “Empirical recurrence of order 90” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 328$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{90} = 64 \neq 0$, so no further tail satisfies anything shorter; any order-90 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A231995 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no element having a strict majority of its horizontal and antidiagonal neighbors equal to one.”. It has offset 1 and begins

168, 3916, 155496, 6257919, 245403000, 9716967198, 383555570610, 15154710590391, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:46:50, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 328$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 328$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 90: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 90.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 656$ exact terms from $n = 2$ on, it returns order 90 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	24	c_{24}	−196470816797080853716	c_{47}	−3580682987119769728858648	c_{70}	14
c_2	1062	c_{25}	363544508791049215603	c_{48}	−8465438000516735893633026	c_{71}	−1
c_3	−11169	c_{26}	1364907780576538524715	c_{49}	2863937260014632176740849	c_{72}	−1
c_4	−352173	c_{27}	−2149208411427681572691	c_{50}	6346311264864238343937990	c_{73}	7
c_5	2439940	c_{28}	−7731647459091792847342	c_{51}	−1923947853142391825085075	c_{74}	9
c_6	57932087	c_{29}	10347971421382291560927	c_{52}	−3985118818266635477432638	c_{75}	−
c_7	−321414705	c_{30}	35911885616746353846504	c_{53}	1083287365003700905772895	c_{76}	−
c_8	−5548184805	c_{31}	−40898398908240568359341	c_{54}	2094481935916977394214711	c_{77}	−
c_9	27155151272	c_{32}	−137424081339079019369989	c_{55}	−509862748637475932337087	c_{78}	−
c_{10}	337618776362	c_{33}	133574848725485737678898	c_{56}	−920174016384896988516788	c_{79}	−
c_{11}	−1507860341417	c_{34}	435002837393182744145292	c_{57}	199938844736514928733361	c_{80}	−
c_{12}	−14002348482395	c_{35}	−362523040472497265690128	c_{58}	337284271123987383010398	c_{81}	−
c_{13}	57202789736145	c_{36}	−1142855163845943313689393	c_{59}	−65065233297058604188401	c_{82}	−
c_{14}	417660786642383	c_{37}	821342465572614206969462	c_{60}	−102873865700492691029570	c_{83}	−
c_{15}	−1544529471945958	c_{38}	2499012178106015082458222	c_{61}	17487136697317988102082	c_{84}	−
c_{16}	−9315116244904336	c_{39}	−1558999720681311538277920	c_{62}	26016506238051669657406	c_{85}	−
c_{17}	30733029336148934	c_{40}	−4558222618603683082063379	c_{63}	−3858920556032269911572	c_{86}	−
c_{18}	159677242566464487	c_{41}	2485599237520199283731141	c_{64}	−5429864055848619935119	c_{87}	−
c_{19}	−463228976094479102	c_{42}	6947496377870008664885875	c_{65}	694204169314794314164	c_{88}	−
c_{20}	−2145130447158653967	c_{43}	−3334217405287797106443800	c_{66}	929560559765425148102	c_{89}	−
c_{21}	5403712782855254339	c_{44}	−8859826151807593618864349	c_{67}	−100920298778349844678	c_{90}	−
c_{22}	22905674483474899818	c_{45}	3765767480936932153824753	c_{68}	−129512211135382634348		
c_{23}	−49618578713051564738	c_{46}	9461574247660911738944221	c_{69}	11729255841815253993		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A231995 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{90} - c_1x^{90-1} - \dots - c_{90}$. Its constant term is $-c_{90} = -64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 90 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 90, so $\deg q = 90$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 90, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 90, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 64.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-90 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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